

Continuing Simple Linear Regression

STAT 223 – Applied Analytics

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Correlation

Correlation describes the *linear* association or dependence between two variables.

A **positive correlation** between two variables means that as one variable increases, the other variable increases as well.

A **negative correlation** between two variables means that as one variable increases, the other variable decreases.

The strength of correlation between a predictor variable and a response variable can be measured by the **correlation coefficient**. The population correlation coefficient is denoted by ρ and the sample correlation coefficient is denoted by R .

Sample Correlation Coefficient, R

The **sample correlation coefficient**, R , can be calculated by:

$$R = \frac{n \sum xy - (\sum x)(\sum y)}{\sqrt{n(\sum x^2) - (\sum x)^2} \sqrt{n(\sum y^2) - (\sum y)^2}}$$

Value of $ R $	Strength of Correlation
$0 < R \leq 0.40$	Weak
$0.40 < R \leq 0.80$	Moderate
$0.80 < R \leq 1.00$	Strong

Hypothesis Test for Correlation

The hypotheses of a correlation test are:

$$H_0 : \rho = 0$$

$$H_A : \rho > 0, \rho < 0 \text{ or } \rho \neq 0$$

It can be shown that, for values of ρ close to 0:

$$R \sim \text{NORMAL} \left(\rho, \frac{1 - \rho^2}{n - 2} \right)$$

Then, the test statistic is:

$$t = \frac{R\sqrt{n-2}}{\sqrt{1-R^2}} \sim t_{n-2}$$

To do this in Python, we can use `scipy.stats.linregress` or `scipy.stats.pearsonr`.

Correlation

The **coefficient of determination**, denoted by R^2 , gives the ratio of the variance in the response variable explained by the predictor variable.

$$R^2 = \frac{\sum (\hat{Y}_i - \bar{Y})^2}{\sum (Y_i - \bar{Y})^2},$$

where $\hat{Y}_i$ is the estimated value of Y_i and $\bar{Y}$ is the mean of the Y_i 's.

Interpretation of the Linear Model

y -Intercept, b_0

The y -intercept, b_0 , is the expected value of Y which $X = 0$. This may not always be a *good* prediction, if our data is not close to 0, but that is the interpretation.

Slope, b_1

If we increase x by one unit, we would expect $\hat{Y}$ to change by the value of the slope, b_1 units. If b_1 is positive, $\hat{Y}$ will increase. If b_1 is negative, $\hat{Y}$ will decrease.

Residual Statistics

Residual Sum of Squares

The **residual sum of squares** is the sum of squared residuals for the sample,

$$SSE = \sum_{i=1}^n (Y_i - \hat{Y}_i)^2.$$

Residual Degrees of Freedom

The **residual degrees of freedom** is $n - p$.

Residual Mean Squared Error

The **residual mean squared error** is the residual sum of squares divided by the residual degrees of freedom,

$$MSE = \frac{SSE}{n - p} = \frac{\sum_{i=1}^n (Y_i - \hat{Y}_i)^2}{n - p}.$$

Residual Standard Error

The **residual standard error** is the square root of the residual mean squared error, $s = \sqrt{MSE}$.

Confidence Intervals for the Predictions

Confidence Interval for the Mean $E(Y|X = x^*)$

The $(1 - \alpha)\%$ confidence interval for the mean $E(Y|X = x^*)$ is given by

$$\hat{Y} \pm t_{n-2, 1-\frac{\alpha}{2}}^* s_y \sqrt{\frac{1}{n} + \frac{(x^* - \bar{x})^2}{(n-1)s_x^2}}$$

Confidence Intervals for the Predictions

Confidence Interval for the Individual Response

The $(1 - \alpha)\%$ confidence interval for the mean $E(Y|X = x^*)$ is given by

$$\hat{Y} \pm t_{n-2, 1-\frac{\alpha}{2}}^* s_y \sqrt{1 + \frac{1}{n} + \frac{(x^* - \bar{x})^2}{(n-1)s_x^2}}$$